Marl Jonson

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Education

San Jose State University **Expected Graduation: May 2027**

Bachelor of Science in Computer Science *San Jose, CA*

- **Coursework:** Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Software Engineering
- **Organizations:** Spartan Analytics, Software & Computer Engineering Society, Responsible Computing Club, Product Management Club

Experience

Nevada National Security Sites May 2026 – Present

Software Engineer Intern *North Las Vegas, NV*

- Kickstarted the creation of a reusable Blazor app template, utility package, and internal wiki using C# and ASP.NET Core Blazor, reducing boilerplate code across 35 apps by centralizing functional components shared between isolated projects into a unified codebase
- Started and completed the independent initiative of migrating 30 apps from .NET 8.x to 10 and from NET Framework 4.x to 4.8 by upgrading NuGet packages/dependencies and resolving build issues on Visual Studio, eliminating urgent EOL technical debt for 12 devs
- Refactored legacy AutoMapper object mappings to Mapster across 25 .NET apps, tracked change requests using ServiceNow, and executed weekly prod deployments after hours to an Azure DevOps pipeline, eliminating $1,500/year in projected commercial licensing liabilities
- Contributed to Agile sprint planning, tracked Azure DevOps work items and user stories, and presented progress during weekly sprints

San Francisco Bay Area Rapid Transit District Apr. 2025 – Apr. 2026

Product Management Intern *Oakland, CA*

- Developed Transiteers, a modular Scratch-based video game introducing youth audiences to public transit careers and workforce pathways
- Interviewed 10+ subject matter experts to translate employee duties, responsibilities, and career insights into interactive learning modules
- Conducted on-site gameplay testing with diverse age demographics across Oakland and Berkeley to evaluate game usability/engagement
- Collected qualitative feedback from users and stakeholders to iteratively redesign game modules, improve flows, and refine feature scope

SJSU Responsible Computing Club Jun. 2025 – Mar. 2026

Software Engineering Ambassador *San Jose, CA*

- Developed and deployed a full-stack web application using React, TypeScript, and backend APIs with version control via GitHub
- Built and maintained responsive web designs using Next.js, Tailwind CSS for styling, and Auth0 for secure authentication
- Created Figma wireframes and user flows for UI/UX prototyping; engaged in bi-monthly Agile sprints and weekly stand-ups
- Performed 5 user interviews using A/B testing to evaluate the usefulness of website features for prospective club members

Projects

Zinnia (HPC Cluster Operations Simulator) | *C++20, C#, .NET 10, ASP.NET Core, gRPC, PostgreSQL, SignalR, React/TypeScript, Docker Compose*

- Built a C++20 event-driven cluster simulator modeling nodes, jobs, queues, failures, retries, contention, and scheduler state
- Developed a .NET 10 control-plane API with gRPC boundaries to manage jobs, node health, metrics, events, and simulation state
- Created a React live dashboard with SignalR to visualize queue depth, utilization, failures, event streams, and scheduling decisions
- Containerized the simulator, API, dashboard, PostgreSQL, Prometheus, and Grafana with Docker Compose and GitHub Actions CI

Shortcake (Power Signals Anomaly Detection Simulator) | *Python, C++20, PyTorch, FastAPI, PostgreSQL, React/TypeScript, Docker Compose, Git*

- Built a C++20 simulation core modeling grid sensors, load changes, outages, noisy telemetry, and infrastructure failure scenarios
- Trained Python anomaly detection models on simulated power-signal data to identify voltage spikes, sensor drift, and outage patterns
- Developed FastAPI services and PostgreSQL models to run experiments, store signal history, and compare detection performance
- Created a React/TypeScript dashboard to visualize telemetry streams, anomaly alerts, failure timelines, and model evaluation metrics

Pollinate (Distributed Event Platform) | *Java, Spring Boot, Kafka, PostgreSQL, Redis, Docker, Kubernetes, OpenTelemetry, Prometheus*

- Built 8 Spring Boot microservices simulating medical devices, exchanging HL7 FHIR events through Apache Kafka
- Implemented retry queues, dead letter queues, idempotent consumers, and distributed transactions for reliable event delivery
- Secured REST and gRPC APIs with OAuth2, JWT authentication, RBAC, and audit logging across distributed services
- Deployed to Kubernetes with Helm, GitHub Actions, OpenTelemetry, Prometheus, and Grafana for CI/CD and observability

Technical Skills

Languages: Java, Python, C#, SQL, TypeScript, Kotlin (novice), Go (novice)

Frameworks & Technologies: Spring Boot, ASP.NET Core, Blazor, React, Next.js, FastAPI, PySpark, Apache Spark, Delta Lake, Databricks, Kafka, PostgreSQL, SQL Server, Redis, Docker, Kubernetes

Developer Tools: Microsoft Azure, Azure DevOps, AWS (novice), GitHub Actions, Terraform, Git, OpenTelemetry, Prometheus, Grafana, REST, gRPC, OAuth2, JWT, Postman, Visual Studio

Certifications: Azure Data Fundamentals (DP-900), Azure Fundamentals (AZ-900), Microsoft 365 Fundamentals (MS-900)